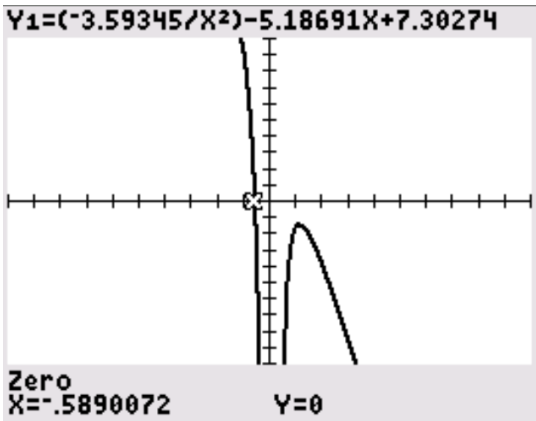
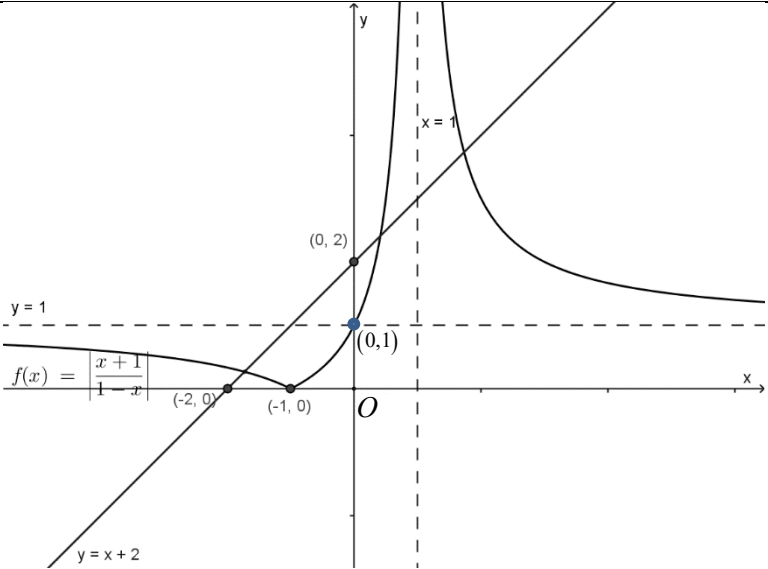
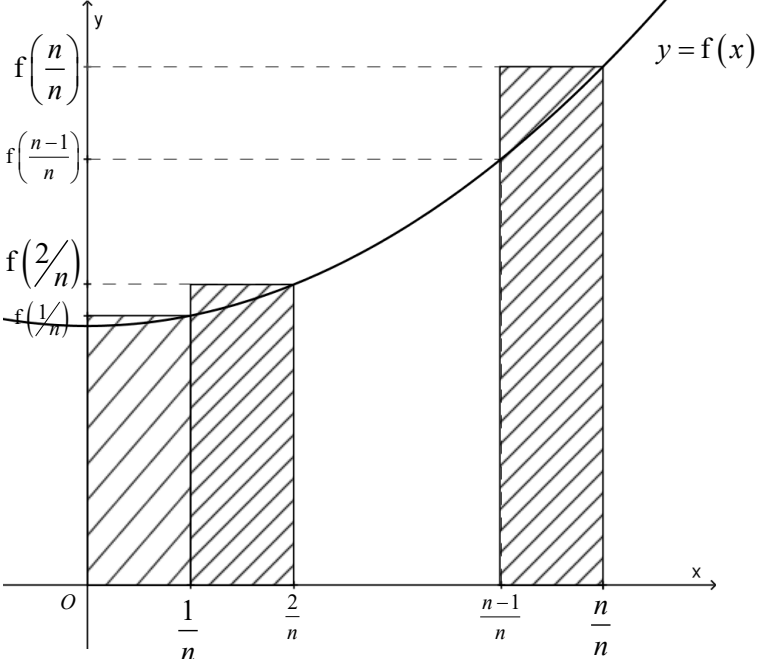
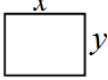
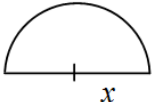
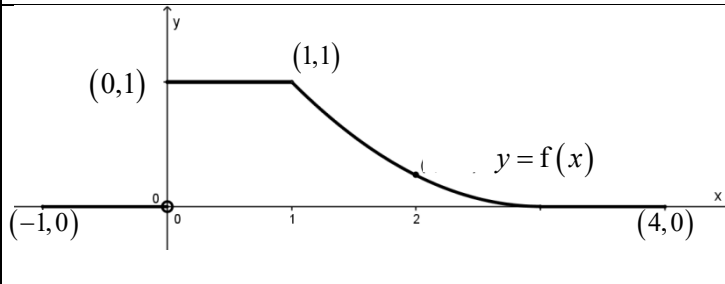
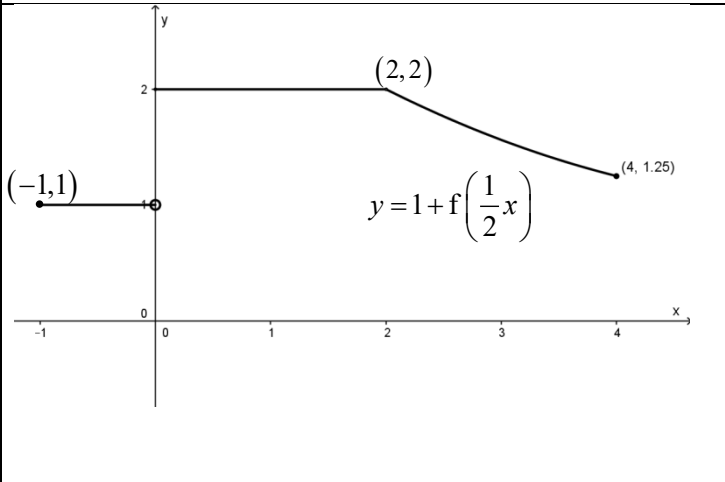
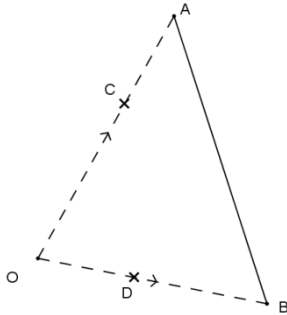


Qn	Solutions	Comments
1i	$y = \frac{a}{x^2} + bx + c$ $-2.4 = \frac{a}{(1.6)^2} + b(1.6) + c \quad \text{--- (1)}$ $3.6 = \frac{a}{(-0.7)^2} + b(-0.7) + c \quad \text{--- (2)}$ $\frac{dy}{dx} = \frac{-2a}{x^3} + b$ $2 = -2a + b \quad \text{--- (3)}$ $a = -3.593$ $b = -5.187$ $c = 7.303 \quad (3.d.p)$	Read question carefully, the degree of accuracy required by the question is 3 decimal places.
ii	$y = \frac{-3.59345}{x^2} - 5.18691x + 7.30274$ $y = 0 \Rightarrow 0 = \frac{-3.59345}{x^2} - 5.18691x + 7.30274$ $5.18691x^3 - 7.30274x^2 + 3.59345 = 0$ By GC, $x = -0.589$ Alternatively, $y = \frac{-3.59345}{x^2} - 5.18691x + 7.30274$  $x = -0.589$	For intermediate working, use at least 2 more significant figures than the required answer. For this question, we also require answers correct to 3 decimal places. We can use the GC app to solve the cubic equation, since exact answers are not required. If you are using the graphical approach, please sketch the graph as your working.
iii	Equation of the other asymptote is $y = -5.187x + 7.303$	Recall how to obtain equations of asymptotes from the form $y = px + q + \frac{r}{x-s}$. Leave your answer as equations. $-5.187x + 7.303$ is not accepted.
2i	$y = \left \frac{x+1}{1-x} \right = \left -1 + \frac{2}{1-x} \right $	Similar to Q1, rewriting as the equation as $y = \left -1 + \frac{2}{1-x} \right$ makes it easier for us to find

Qn	Solutions	Comments
	 Graph of $f(x) = \frac{x+1}{1-x}$ and its asymptotes. The curve has a vertical asymptote at $x=1$ and a horizontal asymptote at $y=1$. It passes through points $(-2, 0)$, $(-1, 0)$, $(0, 1)$, and $(0, 2)$. A line $y=x+2$ is also shown.	the equation of the asymptotes. Main features of curve sketching (Stationary points, Axis Intercepts, Equations of Asymptotes) need to be shown clearly.
ii	By GC, the two graphs intersect at $x = -1.73, 0.414, 1.73$ From the graph, $-1.73 < x < 0.414$ or $x > 1.73$.	You are required to use (i) to solve this question. We are looking for regions where $y = \frac{x+1}{1-x}$ (curve) is below $y = x+2$ (line).
3i	 Area of 1st rectangle $= \frac{1}{n} f\left(\frac{1}{n}\right)$ Area of 2nd rectangle $= \frac{1}{n} f\left(\frac{2}{n}\right)$ $\vdots$ Area of n^{th} rectangle $= \frac{1}{n} f\left(\frac{n}{n}\right)$	One way to start this question is to look at $\frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right\}$ as $\frac{1}{n} f\left(\frac{1}{n}\right) + \frac{1}{n} f\left(\frac{2}{n}\right) + \dots + \frac{1}{n} f\left(\frac{n}{n}\right)$ and compare with $\int_0^1 f(x) dx$, which represents area bounded by graph of $y = f(x)$ from $x = 0$ to $x = 1$. There are tutorial questions and lecture examples on this. “with the aid of a sketch”, means you actually have to draw a sketch for the graph of $y = f(x)$.

Qn	Solutions	Comments
	Sum of area of the n rectangles $= \frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots f\left(\frac{n}{n}\right) \right\}$ As $n \rightarrow \infty$, Sum of area of rectangles $\rightarrow$ area under the graph from $x = 0$ to $x = 1$. $\therefore \lim_{n \rightarrow \infty} \frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots f\left(\frac{n}{n}\right) \right\} = \int_0^1 f(x) \, dx$	
ii	Observe $f(x) = \sqrt[3]{x}$ $\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ \frac{\sqrt[3]{1} + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{\sqrt[3]{n}} \right\} = \int_0^1 \sqrt[3]{x} \, dx$ $= \left[\frac{3}{4} x^{\frac{4}{3}} \right]_0^1$ $= \frac{3}{4}$	For this question, we need to compare $f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots f\left(\frac{n}{n}\right) \text{ and } \frac{\sqrt[3]{1} + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{\sqrt[3]{n}}$ $= \sqrt[3]{\frac{1}{n}} + \sqrt[3]{\frac{2}{n}} + \dots \sqrt[3]{\frac{n}{n}}$
4	  $A = xy + \frac{1}{2}\pi x^2$ $2(x + y) + 2x + \frac{1}{2}\pi(2x) = d$ $2y + (4 + \pi)x = d$ $y = \frac{d - (4 + \pi)x}{2}$ $\therefore A = x \left(\frac{d - (4 + \pi)x}{2} \right) + \frac{1}{2}\pi x^2$ $= x \left(\frac{d}{2} - 2x \right)$ $\frac{dA}{dx} = 0 \Rightarrow \left(\frac{d}{2} - 2x \right) + x(-2) = 0$ $\Rightarrow \frac{d}{2} - 4x = 0$ $\Rightarrow x = \frac{d}{8}$ $\frac{d^2 A}{dx^2} = -4 < 0 \left(\text{when } x = \frac{d}{8} \right) \Rightarrow \text{Area is a maximum}$	Do not assume that exactly half of the wire is used for each shape.

Qn	Solutions	Comments
	$\therefore A = \frac{d}{8} \left(\frac{d}{2} - 2 \left(\frac{d}{8} \right) \right)$ $= \frac{1}{32} d^2$	Need to prove the area calculated is maximum.
4i	$y = x^2$ Translate in the positive x-direction by 3 units $y = (x-3)^2$ Stretch parallel to y-axis, x-axis invariant, with factor $\frac{1}{4}$ $y = \frac{1}{4}(x-3)^2$ $(3,0)$	The order of the transformations can be reversed.
ii		Give the coordinates of all important points. $(0,0)$ is excluded. When $x = 0$, $y = 1$.
iii		x is replaced by $\frac{1}{2}x$ $\downarrow$ Stretch the graph parallel to the x -axis, factor 2, y -axis invariant y is replaced by $y - 1$ $\downarrow$ Translate the resultant graph 1 unit in the positive y -direction $(2, 0.25)$ from previous part becomes our end point.
6i	$\ln(1+2x) = 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} + \dots$ $= 2x - 2x^2 + \frac{8}{3}x^3 + \dots$	Replace x with $2x$ in the formula list

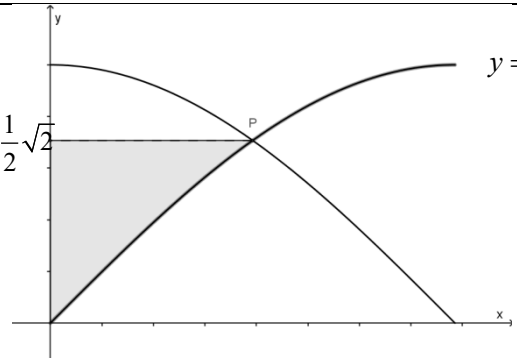
Qn	Solutions	Comments
ii	$ax(1+bx)^c = ax \left[1 + c(bx) + \frac{c(c-1)}{2}(bx)^2 + \dots \right]$ $= ax \left(1 + bcx + \frac{b^2c(c-1)}{2}x^2 + \dots \right)$ $= ax + abcx^2 + \frac{ab^2c(c-1)}{2}x^3 + \dots$ $\therefore a = 2$ $abc = -2 \quad \text{---(1)} \Rightarrow \quad bc = -1$ $\frac{ab^2c(c-1)}{2} = \frac{8}{3} \quad \text{---(2)}$ Substitute (1) into (2) $b(c-1) = \frac{-8}{3}$ $-1-b = \frac{-8}{3}$ $b = -\left(\frac{-8}{3} + 1\right) = \frac{5}{3}$ $c = -\frac{3}{5}$ Term in x^4 for $2x\left(1 + \frac{5}{3}x\right)^{-\frac{3}{5}}$ is $2x \frac{\left(-\frac{3}{5}\right)\left(-\frac{8}{5}\right)\left(-\frac{13}{5}\right)}{3!} \left(\frac{5}{3}x\right)^3 = -\frac{104}{27}x^4$ Coefficient of x^4 is $-\frac{104}{27}$.	
7i	$\overrightarrow{OC} = \frac{3}{5}\underline{a}, \overrightarrow{OD} = \frac{5}{11}\underline{b}$ 	
ii	Since point B is on the line, and $\overrightarrow{BC} = \frac{3}{5}\underline{a} - \underline{b}$ is parallel to the line, equation of line BC is $\underline{r} = \underline{b} + s\left(\frac{3}{5}\underline{a} - \underline{b}\right)$ $\underline{r} = \frac{3}{5}s\underline{a} + (1-s)\underline{b}$	The equation of a line must begin with $\underline{r} = \dots$ and end with $\lambda \in \mathbb{R}$, or in this case, “λ is a parameter.”

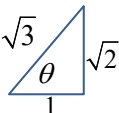
Qn	Solutions	Comments
	Let λ be s, equation of line BC is $\underline{r} = \frac{3}{5}\lambda\underline{a} + (1-\lambda)\underline{b}$, where λ is a parameter. $\overrightarrow{AD} = \frac{5}{11}\underline{b} - \underline{a}$ Equation of line AD is $\underline{r} = \underline{a} + \mu\left(\frac{5}{11}\underline{b} - \underline{a}\right)$ $\underline{r} = (1-\mu)\underline{a} + \frac{5}{11}\mu\underline{b}$, where μ is a parameter.	
iii	When lines AD and BC intersect, $\frac{3}{5}\lambda\underline{a} + (1-\lambda)\underline{b} = (1-\mu)\underline{a} + \frac{5}{11}\mu\underline{b}$ Comparing coefficients of $\underline{a}$ and $\underline{b}$, $\frac{3}{5}\lambda = 1 - \mu \quad \text{---(1)}$ $1 - \lambda = \frac{5}{11}\mu \quad \text{---(2)}$ Solving (1) and (2): $\mu = \frac{11}{20}$ $\overrightarrow{OE} = \left(1 - \frac{11}{20}\right)\underline{a} + \frac{5}{11}\left(\frac{11}{20}\right)\underline{b} = \frac{9}{20}\underline{a} + \frac{1}{4}\underline{b}$ $\overrightarrow{AE} = -\frac{11}{20}\underline{a} + \frac{1}{4}\underline{b} = \frac{11}{20}\left(\frac{5}{11}\underline{b} - \underline{a}\right)$ $\overrightarrow{AE} = \frac{11}{20}\overrightarrow{AD} \quad \Rightarrow \quad AE : ED = 11 : 9$	For this method to work, by right the question has to mention that a and b are not parallel vectors. Recall when $s\underline{a} + t\underline{b} = p\underline{a} + q\underline{b}$ and $\underline{a}$ is not parallel to $\underline{b}$, then $s = p$ and $t = q$
8i	Total time A took = $\frac{50}{2}(2T + 49(2))$ s $= 50T + 2450 \text{ s}$ $5400 \leq 50T + 2450 \leq 6300$ $59 \leq T \leq 77$ $\therefore \{T \in \mathbb{R} : 59 \leq T \leq 77\}$	“each subsequent lap takes 2 seconds longer” hints at an arithmetic progression with common difference 2 Try to solve this as an inequality, rather than as equations
ii	Total time B took = $\frac{t(1.02^{50} - 1)}{1.02 - 1}$ s $= 50t(1.02^{50} - 1) \text{ s}$ $5400 \leq 50t(1.02^{50} - 1) \leq 6300$ $63.845 \leq t \leq 74.486$ $\therefore \{t \in \mathbb{R} : 63.8 \leq t \leq 74.5\}$	“time for each subsequent lap is 2% more than the for the previous lap” hints at a geometric progression with common ratio 1.02

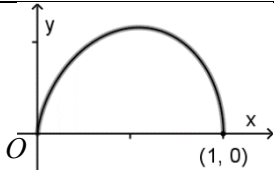
Qn	Solutions	Comments
iii	If A finishes in $1\frac{1}{2}$ hours, timing on 50th lap is $(59 + 49(2)) = 157 \text{ s}$ If B finishes in $1\frac{1}{2}$ hours, timing on 50th lap is $63.845(1.02)^{49} = 168.47\text{s}$ Difference between the timings = $168.47 - 157 = 11.47\text{s}$ $= 11\text{s}$	Final degree of accuracy required is to the nearest second, hence for intermediate working we need at least 2 more decimal places where applicable.
9a	$\frac{w^2}{w^*} = \frac{w^3}{ w ^2}$ $= \frac{(a+ib)^3}{a^2+b^2}$ $= \frac{a^3 - 3b^2a + i(3a^2b - b^3)}{a^2+b^2}$ Since $\frac{w^2}{w^*}$ is purely imaginary, $a^3 - 3b^2a = 0$ $a(a - \sqrt{3}b)(a + \sqrt{3}b) = 0$ $a = 0 \text{ (NA } \because a \neq 0) \text{ or } \sqrt{3}b \text{ or } -\sqrt{3}b$ $b = \frac{1}{\sqrt{3}}a \text{ or } -\frac{1}{\sqrt{3}}a$ Hence, $w = a + i\frac{1}{\sqrt{3}}a$ or $a - i\frac{1}{\sqrt{3}}a$	Final answer is required in terms of a, may not be a good idea to attempt this problem in the polar form, although it can be done. Let $w = re^{i\theta}$ $\frac{w^2}{w^*} = \frac{r^2 e^{2\theta i}}{r e^{-i\theta}} = r e^{3\theta i}$ Hint: when $\frac{w^2}{w^*}$ is purely imaginary, what can we say about r or θ? What can we conclude about a and b?
bi	$z^5 = -32i$ $= 32e^{\left(-\frac{\pi}{2} + 2k\pi\right)i}, k \in \mathbb{Z}$ $z = 2e^{\left(-\frac{\pi + 4k\pi}{10}\right)i}, k = 0, \pm 1, \pm 2$ Hence, $z = 2$ and $\arg(z) = -\frac{\pi}{10}, \frac{3\pi}{10}, -\frac{\pi}{2}, \frac{7\pi}{10}, -\frac{9\pi}{10}$	Not in H2 Math syllabus The question is phrased in a slightly different way from what you might be used to seeing. We will answer according to the requirements of the question, by stating the modulus and arguments. Remember to consider the principal range for arguments.
ii	$z_1 = 2e^{\frac{7\pi}{10}i}, z_2 = 2e^{-\frac{9\pi}{10}i}$ Method 1: $z_1 - z_2$ $= 2 \left[\cos \frac{7\pi}{10} + i \sin \frac{7\pi}{10} - \left(\cos \left(-\frac{9\pi}{10} \right) + i \sin \left(-\frac{9\pi}{10} \right) \right) \right]$	Not in H2 Math syllabus This method uses a lot of trigo. Most of the formulae used can be found in MF15.

Qn	Solutions	Comments
	$= 2 \left[\cos \frac{7\pi}{10} - \cos \left(-\frac{9\pi}{10} \right) + i \left(\sin \frac{7\pi}{10} - \sin \left(-\frac{9\pi}{10} \right) \right) \right]$ $= 2 \left[-2 \sin \left(\frac{\frac{7\pi}{10} + \left(-\frac{9\pi}{10} \right)}{2} \right) \sin \left(\frac{\frac{7\pi}{10} - \left(-\frac{9\pi}{10} \right)}{2} \right) \right.$ $\quad \left. + i \left(2 \cos \left(\frac{\frac{7\pi}{10} + \left(-\frac{9\pi}{10} \right)}{2} \right) \sin \left(\frac{\frac{7\pi}{10} - \left(-\frac{9\pi}{10} \right)}{2} \right) \right) \right]$ $= -4 \sin \left(-\frac{\pi}{10} \right) \sin \left(\frac{4\pi}{5} \right) + 4i \cos \left(-\frac{\pi}{10} \right) \sin \left(\frac{4\pi}{5} \right)$ $= 4 \sin \frac{4\pi}{5} \left[-\sin \left(-\frac{\pi}{10} \right) + i \cos \left(-\frac{\pi}{10} \right) \right]$ $= 4 \sin \frac{4\pi}{5} \left[\sin \left(\frac{\pi}{10} \right) + i \cos \left(\frac{\pi}{10} \right) \right]$ $= 4 \sin \left(\pi - \frac{4\pi}{5} \right) \left[\cos \left(\frac{\pi}{2} - \frac{\pi}{10} \right) + i \sin \left(\frac{\pi}{2} - \frac{\pi}{10} \right) \right]$ $= 4 \sin \frac{\pi}{5} \left[\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right]$ $\arg(z_1 - z_2) = \frac{2\pi}{5} \text{ and } z_1 - z_2 = 4 \sin \left(\frac{1}{5} \pi \right)$ Method 2: $\begin{matrix} S(z_1 - z_2) \\ P(z_1) \end{matrix}$ $\begin{matrix} \text{Re} \\ Q(z_2) \end{matrix}$ Let P, Q, R and S be the points representing the complex numbers $z_1, z_2, -z_2$ and $z_1 - z_2$ respectively. $\arg(-z_2) = -\frac{9\pi}{10} + \pi = \frac{\pi}{10}$ $\angle POR = \frac{7\pi}{10} - \frac{\pi}{10} = \frac{3\pi}{5}$ $\angle ROS = \frac{1}{2} \left(\frac{3\pi}{5} \right) = \frac{3\pi}{10}$ $\arg(z_1 - z_2) = \frac{\pi}{10} + \frac{3\pi}{10} = \frac{2\pi}{5}$	We have to change answer to $\cos \theta + i \sin \theta$ in order to identify the argument. We have to change $\sin \left(\frac{4\pi}{5} \right)$ to $\sin \left(\frac{\pi}{5} \right)$, due to requirement in the question. For this method, we are making use of the parallelogram law of vectors addition for $\overrightarrow{OP} + \overrightarrow{OR}$. We will make use of the properties of a parallelogram to help us solve this question. It is a bit more difficult to use sine rule for this question.

Qn	Solutions	Comments
	$\angle ORS = \pi - \frac{3\pi}{5} = \frac{2\pi}{5}$ $OS = \sqrt{2^2 + 2^2 - 2(2)(2)\cos\left(\frac{2\pi}{5}\right)}$ $= \sqrt{8 - 8\cos\left(\frac{2\pi}{5}\right)}$ $= \sqrt{8 - 8\left(1 - 2\sin^2\frac{\pi}{5}\right)}$ $= \sqrt{16\sin^2\frac{\pi}{5}}$ $= 4\sin\frac{\pi}{5}$	
10i	$A_1 = \int_0^{\frac{\pi}{4}} \sin x \, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x \, dx$ $= [-\cos x]_0^{\frac{\pi}{4}} + [\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ $= -\frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}}$ $= 2 - \frac{2}{\sqrt{2}}$ $A_1 + A_2 = \int_0^{\frac{\pi}{2}} \cos x \, dx$ $A_2 = [\sin x]_0^{\frac{\pi}{2}} - 2 + \frac{2}{\sqrt{2}}$ $= \frac{2}{\sqrt{2}} - 1$ $\frac{A_1}{A_2} = \frac{2 - \frac{2}{\sqrt{2}}}{\frac{2}{\sqrt{2}} - 1}$ $= \frac{2\sqrt{2} - 2}{2 - \sqrt{2}}$ $= \frac{\sqrt{2}(2 - \sqrt{2})}{2 - \sqrt{2}}$ $= \sqrt{2}$	We have to find both areas exactly in order to show that the ratio is $\sqrt{2}$ Alternative Method $A_2 = \int_0^{\frac{\pi}{4}} \cos x \, dx - \int_0^{\frac{\pi}{4}} \sin x \, dx$ $= [\sin x]_0^{\frac{\pi}{4}} + \left(-\frac{1}{\sqrt{2}} + 1\right)$ $= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1$ $= \frac{2}{\sqrt{2}} - 1$

Qn	Solutions	Comments
ii	 $y = \sin x \Rightarrow x = \sin^{-1} y$ $\text{Vol} = \int_0^{\frac{\sqrt{2}}{2}} \pi x^2 dy$ $= \pi \int_0^{\frac{\sqrt{2}}{2}} (\sin^{-1} y)^2 dy$	Avoid skipping steps for a “show” question
iii	$\pi \int_0^{\frac{\sqrt{2}}{2}} (\sin^{-1} y)^2 dy$ $= \pi \int_0^{\frac{\pi}{4}} u^2 (\cos u) du$ $= \pi \left(\left[u^2 \sin u \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} 2u \sin u du \right)$ $= \pi \left(\frac{\pi^2}{16} \left(\frac{\sqrt{2}}{2} \right) - 2 \left(\left[-u \cos u \right]_0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} \cos u du \right) \right)$ $= \pi \left(\frac{\sqrt{2}\pi^2}{32} - 2 \left(\left(-\frac{\pi}{4} \right) \left(\frac{\sqrt{2}}{2} \right) + \left[\sin u \right]_0^{\frac{\pi}{4}} \right) \right)$ $= \pi \left(\frac{\sqrt{2}\pi^2}{32} - 2 \left(-\frac{\sqrt{2}\pi}{8} + \frac{\sqrt{2}}{2} \right) \right)$ $= \frac{\sqrt{2}\pi^3}{32} + \frac{\sqrt{2}\pi^2}{4} - \sqrt{2}\pi \text{ units}^3$ Let $y = \sin u$ $\frac{dy}{du} = \cos u$ When $y = 0, u = 0$ When $y = \frac{\sqrt{2}}{2}, u = \frac{\pi}{4}$	Be very careful when simplifying your integration by parts. Use brackets to help avoid careless mistakes.
11i	$x = \sin^3 \theta$ $y = 3 \sin^2 \theta \cos \theta$ $\frac{dx}{d\theta} = 3 \sin^2 \theta \cos \theta$ $\frac{dy}{d\theta} = 6 \sin \theta \cos^2 \theta - 3 \sin^3 \theta$ $\frac{dy}{dx} = \frac{6 \sin \theta \cos^2 \theta - 3 \sin^3 \theta}{3 \sin^2 \theta \cos \theta}$ $= 2 \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$ $= 2 \cot \theta - \tan \theta$	
ii	When $\frac{dy}{dx} = 0, 2 \cot \theta - \tan \theta = 0$	

Qn	Solutions	Comments																
	$2 - \tan^2 \theta = 0$ $\tan \theta = \pm \sqrt{2} \left(\text{reject } -\sqrt{2} \because 0 \leq \theta \leq \frac{\pi}{2} \right)$ $\tan \theta = \sqrt{2}$	You have to explain why the negative root is rejected.																
	$x = \sin^3 \theta$ $= \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3$ $= \frac{2\sqrt{2}}{3\sqrt{3}}$ $y = 3 \sin^2 \theta \cos \theta$ $= 3 \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^2 \left(\frac{1}{\sqrt{3}} \right)$ $= \frac{2}{\sqrt{3}}$ Coordinates of the turning point: $\left(\frac{2\sqrt{2}}{3\sqrt{3}}, \frac{2}{\sqrt{3}} \right)$ To show maximum: Method 1: To determine that it is a max turning point, use first derivative sign test. <table><tr><td>θ</td><td>$\theta = \sqrt[3]{\sin 0.5}$ ≈ 0.917</td><td>$\theta = \tan^{-1} \sqrt{2}$</td><td>$\theta = \sqrt[3]{\sin 0.6}$ ≈ 1.00</td></tr><tr><td>x</td><td>$x = 0.5$</td><td>$x = \frac{2\sqrt{2}}{3\sqrt{3}} \approx 0.544$</td><td>$x = 0.6$</td></tr><tr><td>$\frac{dy}{dx}$</td><td>0.228</td><td>0</td><td>-0.296</td></tr><tr><td>sign of $\frac{dy}{dx}$</td><td>Positive</td><td>Zero</td><td>Negative</td></tr></table> As seen from the table above, stationary point is a maximum.	θ	$\theta = \sqrt[3]{\sin 0.5}$ ≈ 0.917	$\theta = \tan^{-1} \sqrt{2}$	$\theta = \sqrt[3]{\sin 0.6}$ ≈ 1.00	x	$x = 0.5$	$x = \frac{2\sqrt{2}}{3\sqrt{3}} \approx 0.544$	$x = 0.6$	$\frac{dy}{dx}$	0.228	0	-0.296	sign of $\frac{dy}{dx}$	Positive	Zero	Negative	 $\sin \theta = \frac{\sqrt{2}}{\sqrt{3}}$ $\cos \theta = \frac{1}{\sqrt{3}}$ This part on finding the coordinates can be a bit more challenging, since θ is not one of the special angles when $\tan \theta = \sqrt{2}$. The thing about trigo ratios is that they are obtained from right angled triangles. Using the idea of right angled triangles, we can actually use Pythagoras theorem and TOA CAH SOH to find the required values. To show that it is a maximum point, we use the 1st derivative test. <u>This test is done by testing the values of $\frac{dy}{dx}$ at the neighbourhood of x not θ.</u> You should also calculate the values and not simply put +, -. Method 2 uses the 2nd derivative test. We are actually applying implicit differentiation when we differentiate $\frac{dy}{dx}$ wrt θ.
θ	$\theta = \sqrt[3]{\sin 0.5}$ ≈ 0.917	$\theta = \tan^{-1} \sqrt{2}$	$\theta = \sqrt[3]{\sin 0.6}$ ≈ 1.00															
x	$x = 0.5$	$x = \frac{2\sqrt{2}}{3\sqrt{3}} \approx 0.544$	$x = 0.6$															
$\frac{dy}{dx}$	0.228	0	-0.296															
sign of $\frac{dy}{dx}$	Positive	Zero	Negative															

Qn	Solutions	Comments
	Method 2: (Not in the syllabus) $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$ $= \frac{d}{dx} (2 \cot \theta - \tan \theta)$ $= \frac{d}{d\theta} (2 \cot \theta - \tan \theta) \times \left(\frac{d\theta}{dx} \right) \quad \text{Since } 0 \leq \theta \leq \frac{\pi}{2},$ $= (-2 \operatorname{cosec}^2 \theta - \sec^2 \theta) \left(\frac{1}{3 \sin^2 \theta \cos \theta} \right)$ $= -\frac{2 \operatorname{cosec}^2 \theta + \sec^2 \theta}{3 \sin^2 \theta \cos \theta}$ $2 \operatorname{cosec}^2 \theta + \sec^2 \theta > 0$ and $3 \sin^2 \theta \cos \theta > 0$ $\frac{d^2y}{dx^2} = -\frac{2 \operatorname{cosec}^2 \theta + \sec^2 \theta}{3 \sin^2 \theta \cos \theta} < 0$ The turning point is a maximum point.	
iii	Area $= \int_0^1 y \, dx$ $= \int_0^{\frac{\pi}{2}} (3 \sin^2 \theta \cos \theta) (3 \sin^2 \theta \cos \theta) \, d\theta$ $= \int_0^{\frac{\pi}{2}} (3 \sin^2 \theta \cos \theta)^2 \, d\theta$ $= \int_0^{\frac{\pi}{2}} 9 \sin^4 \theta \cos^2 \theta \, d\theta$ $= 0.884 \text{ units}^2$  When $y = 0$, $3 \sin^2 \theta \cos \theta = 0$ $\theta = 0 \text{ or } \frac{\pi}{2}.$ When $\theta = 0$, $x = 0$ When $\theta = \frac{\pi}{2}$, $x = 1$	Avoid skipping steps for a “show” question
iv	$y = ax$ $3 \sin^2 \theta \cos \theta = a \sin^3 \theta$ Since $y \neq 0$ at P, $\frac{\sin^3 \theta}{\sin^2 \theta \cos \theta} = \frac{3}{a}$ $\tan \theta = \frac{3}{a}$ At maximum point, $\tan \theta = \sqrt{2}$ $\frac{3}{a} = \sqrt{2}$ $a = \frac{3}{\sqrt{2}}$	